

Finding Polyphony Within Human and LLM Annotator Subgroups Through Polarized Trees

The PRG Framework

We refer to the structures produced by our framework as **Polarized Trees**: rooted acyclic graphs in which each node represents an annotator subgroup and each edge a demographic split. The term *polarized* reflects both the input condition—only texts with meaningful polarization are analysed—and the output: leaf nodes representing stable, opposing opinion poles across socio-demographic or ideological dimensions. The goal is to identify, using human annotations alone, the specific subgroups that are most polarized for a given text, serving as a diagnostic tool for understanding which dimensions drive disagreement and an attribution framework for pinpointing intersectional subgroups whose opinions diverge most strongly.

Step 1: Compute nDFU

For each text in the dataset, compute the normalized Distance from Unimodality (nDFU) over its full annotation multiset A , yielding $\text{nDFU}(A) \in [0, 1]$.

Step 2: Filter Polarized Texts

Retain only texts whose nDFU exceeds a predefined threshold θ :

$$\mathcal{D}_\theta = \{A : \text{nDFU}(A) \geq \theta\} \quad (1)$$

Only texts in $\mathcal{D}_\theta$ proceed to tree construction. Texts below θ exhibit no meaningful polarization and are discarded.

Step 3: Set Parameters and Choose PRG Variant

Select the splitting criterion and set the tree construction parameters: minimum subgroup size min_size , polarization gain threshold h , and maximum depth max_depth .

Polarization Reduction Gain (PRG). Analogously to Information Gain in decision trees—which measures the reduction in entropy achieved by splitting on a given feature—we define **Polarization Reduction Gain (PRG)** as the reduction in nDFU achieved by partitioning annotations along a candidate dimension. PRG generalizes the original A Posteriori Unimodality (AU) framework from a binary pass/fail filter to a continuous metric: whereas AU required every subgroup to be perfectly unimodal ($\text{nDFU} = 0$), PRG measures the degree of reduction, yielding two key advantages: (i) *quantifiable attribution* — every text receives a score rather than a binary verdict; (ii) *dimension ranking* — computing PRG across all candidate dimensions identifies which one best explains the observed disagreement.

Let A be the annotation multiset and $\{g_1, \dots, g_G\}$ the subgroups produced by partitioning on dimension dim . We consider two variants:

1. $\text{PRG}_{\text{max}}(A, \text{dim}) = |\text{nDFU}_{\text{global}}(A) - \max_k \text{nDFU}(g_k)|$
2. $\text{PRG}_{\text{var}}(A, \text{dim}) = \left| \text{nDFU}_{\text{global}}(A) - \sum_k \frac{N_k}{N} \text{nDFU}(g_k) \right|$

Both variants lie in $[0, 1]$ since $\text{nDFU} \in [0, 1]$. PRG_{max} targets worst-case residual polarization, ensuring every subgroup benefits from the split and serving as the primary splitting criterion. PRG_{var} captures the size-weighted aggregate reduction—the direct analogue of variance reduction in regression trees—and is reported as a complementary diagnostic.

Since $\text{nDFU}_{\text{global}}$ is fixed at each node, maximising PRG reduces to minimising only the residual term. The splitting objective thus becomes:

1. $\arg \min_{\text{dim}} \max_k \text{nDFU}(g_k)$

$$2. \arg \min_{dim} \sum_k \frac{N_k}{N} \text{nDFU}(g_k)$$

This mirrors the role of impurity minimisation in classical decision trees, with the goal of producing the purest—least polarized—subgroups at each split.

To jointly optimise both objectives, we combine PRG_{max} and PRG_{var} into a single composite score via an F-score-style harmonic mean:

$$\text{PRG}_\beta(A, dim) = \frac{(1 + \beta^2) \cdot \text{PRG}_{max}(A, dim) \cdot \text{PRG}_{var}(A, dim)}{\beta^2 \cdot \text{PRG}_{max}(A, dim) + \text{PRG}_{var}(A, dim)}, \quad \beta > 0 \quad (2)$$

The harmonic mean is preferred over a simple weighted average because it *penalises imbalance*: a dimension that scores well on one variant but poorly on the other is pulled down, ensuring the selected split is genuinely beneficial both in the worst case and in aggregate. This mirrors the motivation for F-score over accuracy in classification. The parameter β controls the relative emphasis: $\beta = 1$ weights both variants equally; $\beta > 1$ emphasises PRG_{max} , favouring conservative splits that protect every subgroup; $\beta < 1$ emphasises PRG_{var} , favouring splits with the greatest aggregate polarization reduction.

Step 4: Construct One Polarized Tree per Text

For each text in $\mathcal{D}_\theta$, recursively partition the annotation multiset along candidate dimensions. At each node, select the dimension that optimises the chosen PRG variant — equivalently, the dimension that minimises the residual polarization term, since nDFU_{global} is fixed at each node:

$$best_dim = \arg \text{opt}_{dim \in dims} \text{PRG}(A_{node}, dim) \quad (3)$$

Once used, a dimension is removed from $dims$ for all descendant nodes on that branch. Splitting terminates at a node when any of the following stopping criteria is met:

- *min_size*: a resulting subgroup would fall below the minimum size, making nDFU estimates unreliable.
- *h*: the best available PRG does not exceed the gain threshold, meaning no dimension meaningfully reduces polarization.
- *max_depth*: the depth limit is reached.
- *Dimension exhaustion*: all candidate dimensions have been used on the current branch.

Minimum Annotator Requirement. For a split along dimension dim to be valid, every resulting subgroup must contain at least $n_{min} = 2$ annotations:

$$\min_{g \in \text{groups}(dim)} |A_g| \geq 2 \quad (4)$$

This requirement is grounded in the structure of the subgroup histogram underlying DFU: a subgroup with a single annotation is trivially unimodal regardless of the label assigned, rendering DFU uninformative. With at least 2 annotations, the subgroup histogram can express disagreement across the label space, making DFU well-defined. In practice, *min_size* enforces this constraint locally at each node during tree construction.

Algorithm. *Notation.* A : annotation multiset; $dims$: candidate dimensions (updated per branch); $\text{groups}(dim, X)$: partition of X by dim ; $\text{labels}(X)$: label multiset; $\text{mode}(L)$: majority label.

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1: function DETECTPOLARIZEDSUBGROUPS( $A, dims, min\_size, h, max\_depth$ )
2:   return DFS( $A, dims, 1$ )
3: function DFS( $node, dims, depth$ )
4:   if  $depth > max\_depth$  or  $dims = \emptyset$  then
5:     return  $\{node\}$ 
6:    $best\_dim \leftarrow \text{null}$ ;  $best\_PRG \leftarrow 0$ 
7:   for  $dim \in dims$  do
8:     if  $\forall g \in \text{groups}(dim, node) : |g| \geq min\_size$  then
9:        $PRG \leftarrow \text{PRG}(node, dim)$ 
10:      if  $PRG > best\_PRG$  then
11:         $best\_PRG \leftarrow PRG$ ;  $best\_dim \leftarrow dim$ 
12:   if  $best\_dim = \text{null}$  or  $best\_PRG \leq h$  then
13:     return  $\{node\}$ 
14:    $dims' \leftarrow dims \setminus \{best\_dim\}$ 
15:   return  $\bigcup_{c \in \text{groups}(best\_dim, node)} \text{DFS}(c, dims', depth + 1)$ 

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Step 5: Assign Pole Labels to Leaf Nodes

For each leaf subgroup g , compute the toxic proportion $p_g = \text{count}(\text{rating} \geq \Theta)/N_g$ and assign:

$$\text{PoleLabel}_g = \begin{cases} \text{toxic} & p_g > 0.5 \\ \text{civil} & p_g < 0.5 \\ \text{indeterminate} & p_g = 0.5 \end{cases} \quad (5)$$

Indeterminate leaves are pruned via Contextual Pruning, yielding the set of determinate leaves $\mathcal{L}^* = \{l \in \mathcal{L} : p_l \neq 0.5\}$, partitioned into $\mathcal{L}_{toxic}^*$ and $\mathcal{L}_{civil}^*$.

Step 6: Aggregate Dataset-Level Metrics

We restrict the following analysis exclusively to trees in $\mathcal{T}$. Let $\mathcal{S}$ be the set of all distinct intersectional subgroups appearing as determinate leaves across $\mathcal{T}$, including those appearing only once. For each $s \in \mathcal{S}$, let $\mathcal{T}_s \subseteq \mathcal{T}$ denote the trees in which s appears as a leaf and $n_s = |\mathcal{T}_s|$ its occurrence count. We define three dataset-level metrics:

1. **Dimension Frequency ($\mathcal{F}$)**. How often each dimension appears as the splitting dimension at depth $d = 1, 2, \dots$ across $\mathcal{T}$. High frequency at $d = 1$ signals a primary polarization driver; at $d > 1$, a secondary intersectional role.
2. **Subgroup Pole Consistency ($\mathcal{C}$)**. For each $s \in \mathcal{S}$:

$$\mathcal{C}(s) = \left(n_s, \frac{|\{t \in \mathcal{T}_s : \text{pole}(s, t) = \text{toxic}\}|}{n_s}, \frac{|\{t \in \mathcal{T}_s : \text{pole}(s, t) = \text{civil}\}|}{n_s} \right) \quad (6)$$

where s is an intersectional subgroup, t a polyphonic tree in $\mathcal{T}_s$, and $\text{pole}(s, t)$ the pole assigned to s in tree t . High n_s with a dominant pole proportion signals stable demographic alignment. Subgroups with $n_s = 1$ are retained as evidence of rare but genuine intersectional structure.

3. **Subgroup PRG ($\mathcal{P}$)**. For each $s \in \mathcal{S}$:

$$\mathcal{P}(s) = \frac{1}{n_s} \sum_{t \in \mathcal{T}_s} \text{PRG}_\beta(s, t) \quad (7)$$

where $\text{PRG}_\beta(s, t)$ is the PRG of the split that produced s in tree t , and the sum averages this value across all n_s trees in which s appears as a leaf. This captures how strongly and consistently the split producing s reduces polarization across the dataset.

$\mathcal{F}$ reveals *where* each dimension drives polarization; $\mathcal{C}$ identifies *who* holds stable opinions; $\mathcal{P}$ quantifies *how strongly* polarization is reduced at each intersectional subgroup.

Robustness of the Metrics These metrics are robust to structural variation across trees. $\mathcal{F}$ uses relative frequencies across $\mathcal{T}$, so trees of different depths contribute proportionally to the counts at each level without conflation. $\mathcal{C}$ operates per subgroup, aggregating only over $\mathcal{T}_s$ —the trees in which s actually appears—so pole proportions are never diluted by trees where s is absent. $\mathcal{P}$ is interpreted as signal strength rather than an absolute score: a low mean PRG indicates that the split producing s was consistently weak, which is itself a meaningful finding. Together, $\mathcal{F}$, $\mathcal{C}$, and $\mathcal{P}$ lift the Polarized Tree from a per-text diagnostic into a corpus-level attribution instrument, revealing which demographic dimensions drive disagreement, which subgroups hold stable opinions, and how strongly each intersectional partition explains the observed polarization.

Working Example

Three dimensions—Gender (male/female), Orientation (heterosexual/lgbtq+), Politics (left/right)—applied to a misogynistic comment ($n = 16$, $\text{nDFU}_{global} = 0.60$). Parameters: $\text{min_size} = 3$, $h = 0.05$, $\text{max_depth} = 5$, $\text{variant} = \text{PRG}_{max}$.

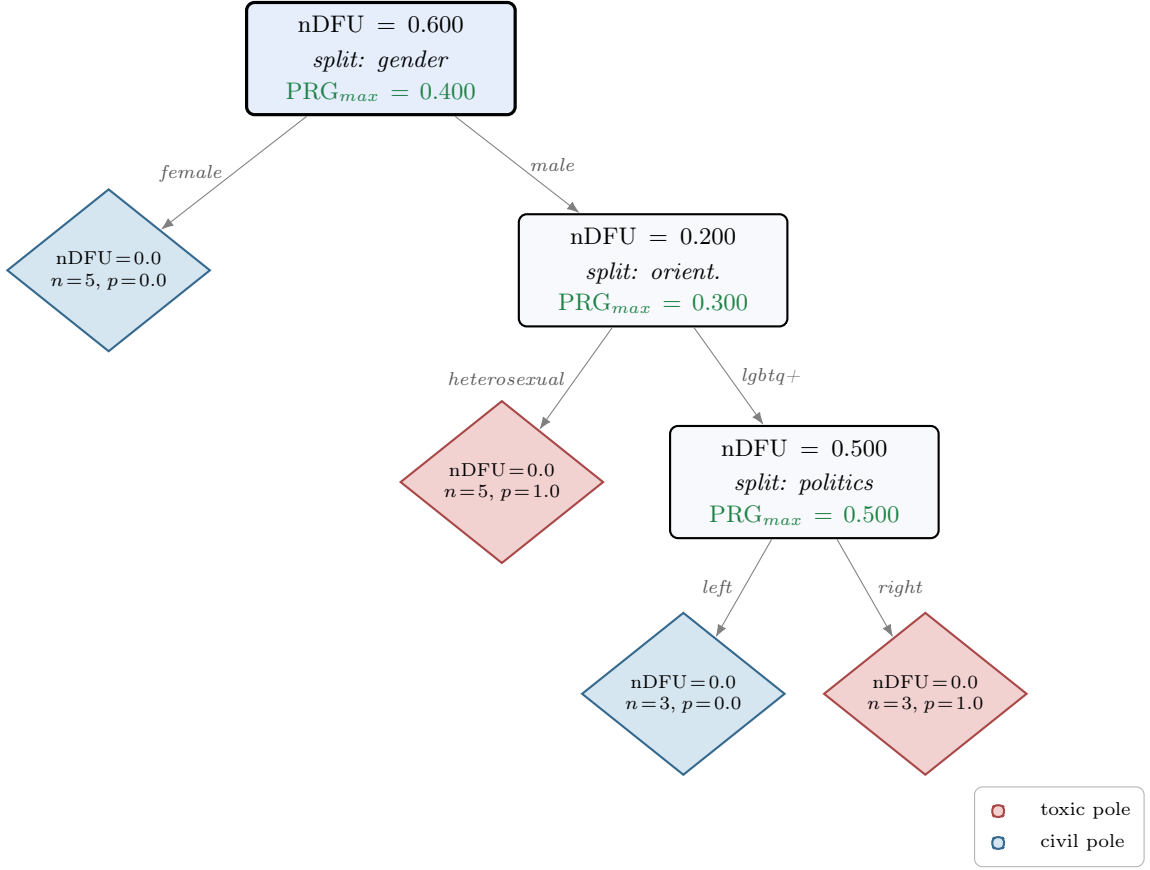


Figure 1: Polarized tree for the misogynistic comment. Rectangles = internal nodes (split dimension + PRG_{max}); diamonds = leaf nodes (red toxic, blue civil).

Table 1: PRG_{max} at each node. SKIP = $\min_size$ not met; $\star$ = dimension selected.

Node	Dim.	$\max_k \text{nDFU}(g_k)$	PRG_{max}	Result
<i>Root</i> ($n = 16$, $\text{nDFU} = 0.60$)				
	Gender	0.200	$ 0.60 - 0.20 = 0.40$	$\star$ split
	Orientation	0.533	$ 0.60 - 0.53 = 0.07$	
	Politics	—	—	SKIP
<i>male</i> ($n = 11$, $\text{nDFU} = 0.20$)				
	Orientation	0.500	$ 0.20 - 0.50 = 0.30$	$\star$ split
	Politics	—	—	SKIP
<i>female</i> ($n = 5$): no dim. passes $\min_size \Rightarrow$ leaf				
<i>male</i> $\rightarrow$ <i>lgbtq+</i> ($n = 6$, $\text{nDFU} = 0.50$)				
	Politics	0.000	$ 0.50 - 0.00 = 0.50$	$\star$ split
<i>male</i> $\rightarrow$ <i>hetero</i> ($n = 5$): Politics skipped $\Rightarrow$ leaf				
<i>lgbtq+</i> $\rightarrow$ <i>left</i> & <i>right</i> ($n = 3$ each): no dims remain $\Rightarrow$ leaves				

Table 2: Pole assignment at leaf nodes ($p_{tox} = \text{count}(\text{rating} \geq 3)/n$).

Leaf subgroup	n	nDFU	p_{tox}	Pole
female	5	0.00	0.00	civil
male $\rightarrow$ heterosexual	5	0.00	1.00	toxic
male $\rightarrow$ lgbtq+ $\rightarrow$ left	3	0.00	0.00	civil
male $\rightarrow$ lgbtq+ $\rightarrow$ right	3	0.00	1.00	toxic

Synthetic Data

To evaluate whether Polarized Trees recovers known polarization structure, we construct a synthetic dataset where disagreement is not random but explicitly injected via annotator identity. Rather than relying on naturalistic annotation—where ground truth polarization structure is unknown—we design a controlled setting in which the generative process knows exactly what drives disagreement for each text, allowing recovery to be measured directly against this generative ground truth. A method that reliably recovers injected structure on synthetic data provides strong evidence that it will detect genuine structure in real data.

The synthetic setup is grounded in a population analogy. We construct a large pool of annotators spanning all possible intersectional identity combinations — the full universe of demographic profiles that could plausibly annotate a text. This pool represents the true, underlying bias structure of the world: a structure that exists but is never directly observable in reality. For each text, only a subset of the pool is sampled to actually produce ratings, mirroring how any real annotation effort only ever observes a partial, noisy glimpse of the broader population.

Annotator Identities

Each synthetic annotator is assigned a fixed demographic identity across five dimensions: **gender** (male, female, non-binary), **political leaning** (left, center, right), **age group** (<25, 25–50, >50), **educational level** (low, medium, high), and **sexual orientation** (heterosexual, lgbtq+). The Cartesian product of these dimensions yields $3 \times 3 \times 3 \times 3 \times 2 = 162$ distinct identity combinations, each corresponding to an intersectional subgroup in the sense of the Polarized Tree framework.

Annotator Pool

We construct the annotator pool by assigning 10 annotators to each of the 162 identity combinations, yielding 1,620 annotators in total. This guarantees that every intersectional subgroup is sufficiently represented for splits to be explored at any depth without violating a `min_size` constraint.

Each annotator is a persistent identity — a fixed demographic profile that exists once in the pool. For every text, a random subset of 150 annotators is sampled from the pool to actually produce ratings. This sampling step is what separates the pool (privileged, full population) from the observed annotations (the partial sample any real method would see).

Bias Generation

Rather than designating a single dominant dimension per text, we generate intersectional structure by assigning every value of every dimension a random weight drawn from (0.5, 2.0), independently for each text. A value’s weight represents the strength of its pull on annotator judgement for that text: higher weights push toward a toxic rating, lower weights push toward civil, relative to the rest of the pool. An annotator’s combined score is the geometric mean of their weights across all five dimensions, so their final rating depends on their full demographic profile rather than any single trait in isolation.

For each text, a threshold is computed as the median combined score across the *entire* 1,620-annotator pool — not just the 150 sampled annotators. The threshold is therefore empirical, not fixed: it reflects wherever the distribution of combined scores actually falls for that text’s particular random weight draw, guaranteeing an approximately even split regardless of how the weights happen to land. This reflects the population analogy directly: the threshold encodes the true, underlying bias structure for that text, which the sampled annotations only partially reveal. Annotators whose combined score exceeds the threshold are assigned to the toxic pole; the remainder to the civil pole.

This weight-based combination produces genuinely multidimensional behavior: no single dimension is privileged by construction, and the dimension or combination of dimensions that best explains the resulting disagreement varies from text to text, exactly as required for a per-text diagnostic tool to have something non-trivial to discover.

We selected (0.5, 2.0) as the weight range based on empirical testing rather than a fixed analytical derivation. Values below 0.5 shrink rapidly when combined across five dimensions via the geometric mean, leaving little room for a given dimension to register a meaningful contribution; extending the upper bound to 2.0 gives enough spread for some dimension values to carry a genuinely significant pull relative to others, rather than all values blending into a narrow, indistinguishable band. We compared this range against the more directly interpretable (0, 1) and observed less stable behavior in the resulting depth-wise nDFU reduction; (0.5, 2.0) was retained as the more empirically robust choice.

Polarization Regimes

Each text is independently assigned one of three regimes, mixed at a 70/15/15 ratio:

- **Polarized.** The weight-based mechanism above is applied with full-strength, non-overlapping toxic and civil rating ranges, producing strong bimodal disagreement.
- **Weakly polarized.** The same mechanism is applied, but with narrower, overlapping rating ranges, producing a softer signal that a tree should resolve in fewer splits or not at all.
- **Unimodal.** No demographic structure is injected. All sampled annotators rate around a single randomly placed peak on the scale (which may sit anywhere, not only at the midpoint), with natural spread and a small probability of fully random dissent. This regime serves as a negative control: no dimension should explain meaningful disagreement, and recovery should correctly find none.

Across all regimes, a small global noise probability allows any sampled annotator’s rating to be drawn uniformly from the full scale, regardless of regime, reflecting the irreducible noise present in real human judgment.

Validation

We validate the generated data using nDFU (Pavlopoulos and Likas, 2023) prior to running the Polarized Tree itself. Overall nDFU is computed per text, alongside the lowest achievable within-group nDFU for every single dimension and for increasing combinations of dimensions (depth 1 through depth 4). Across regimes, polarized texts show high overall nDFU with a strictly monotonic reduction as additional dimensions are considered; weakly polarized texts show a markedly smaller initial signal; and unimodal texts show near-zero overall nDFU regardless of how annotations are partitioned. This three-way separation, together with the consistent depth-wise reduction observed in polarized texts, provides empirical evidence that the synthetic dataset carries the multidimensional structure the method is designed to recover, prior to evaluating the tree algorithm itself.

Limitations

Annotator density. The minimum annotator count grows with the number of dimensions and their cardinalities, excluding most NLP datasets where fewer than 10 raters annotate each text. That is the reason we will generate synthetic data and work with them

Subgroup balance. Demographic imbalance (e.g. far more heterosexual than lgbtq+ annotators) raises the effective annotator requirement beyond the theoretical bound.

Comment-level operation. The optimal splitting dimension varies across texts and cannot be determined dataset-wide without obscuring text-specific structure. The dataset-level metrics in Step 7 address this by aggregating tree outputs post-hoc rather than imposing a global splitting order.